Auto Ticket Classification using Flow Designer: Automated School IT Ticket Triage on ServiceNow

Project Description:

Auto Ticket Classification streamlines IT support in a school or college by automatically categorizing incoming tickets on the ServiceNow platform. The solution includes a Ticket Categorizer, which reads the short description of every new ticket and fills in the Category and Subcategory, and an Email Notification module, which sends the caller a confirmation email once the ticket has been submitted.

Built entirely with ServiceNow Flow Designer, the low-code automation tool of the Now Platform, the flow named **Auto Classify School IT Tickets** triggers when a record is created on the Incident WorkFlow table with an empty Category. It checks the short description against keyword conditions, updates the record and emails the caller, so that the support team receives tickets that are already organised, without any manual triage.

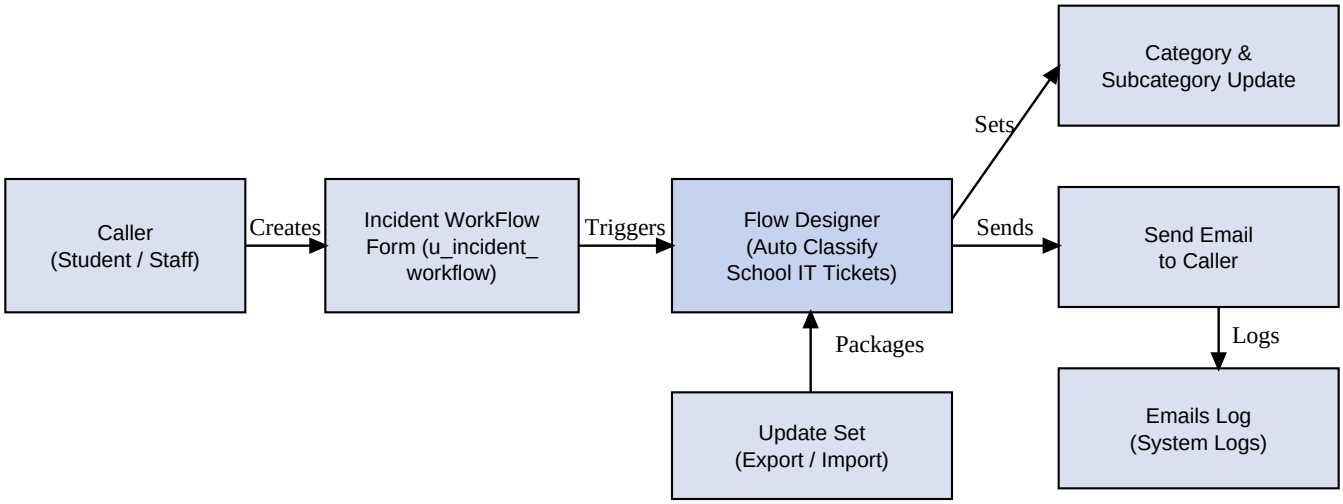
Scenarios

Scenario 1: A user submits a ticket saying "My college has issue with Network". The flow finds the network keyword, sets the Category to Network and the Subcategory to Wi-Fi, and sends a confirmation email to the caller within seconds.

Scenario 2: A staff member reports that the classroom projector is not working. The flow detects the word projector, sets the Category to Hardware and the Subcategory to Projector, and notifies the caller by email.

Scenario 3: A student writes "Forgot password" or "Slow computer". The flow classifies the first as Access / Forgot Password and the second as Performance / Slow Computer, so that each issue reaches the right kind of support.

Technical Architecture:



Description: The solution follows a simple three-layer design on the Now Platform. The presentation layer is the Incident WorkFlow form, where users log a ticket. The automation layer is a Flow Designer flow that triggers on record creation, evaluates If / Else If conditions on the short description, calls Update Record actions for the Category and Subcategory and finally uses a Send Email action to notify the caller. The data layer is the Incident WorkFlow table (u_incident_workflow), which stores the ticket. The sent emails are visible in the Emails log, and the whole solution is packaged in an Update Set so that it can be exported and imported on another instance.

(Note: If you are using a database in your project you definitely need to add an ER Diagram along with a description. This project uses the ServiceNow table Incident WorkFlow (u_incident_workflow) with the standard Caller, Category, Subcategory and Short description fields, so no separate database is created.)

Pre-requisites:

- **ServiceNow Platform Knowledge:** ServiceNow Documentation
- **ServiceNow Developer Instance:** Personal Developer Instance (PDI)
- **Flow Designer / Workflow Studio Familiarity:** Flow Designer Documentation
- **Incident and Table Basics:** ServiceNow Tables
- **Email Notification Basics:** Email Notifications
- **Update Set Handling:** Update Sets

Project Workflow:

Milestone 1: Platform Setup and Architecture

- **Activity 1.1:** Log in to the ServiceNow instance and open Flow Designer.
- **Activity 1.2:** Define the categories, subcategories and keywords used for classification.
- **Activity 1.3:** Define the architecture of the solution.
- **Activity 1.4:** Prepare the environment and the Incident WorkFlow table.

Milestone 2: Core Functionalities Development

- **Activity 2.1:** Develop the core functionalities: Ticket Categorizer and Email Notification.
- **Activity 2.2:** Configure the Category and Subcategory values and the email content.

Milestone 3: Flow Designer Development

- **Activity 3.1:** Create the flow, configure its trigger and add the classification conditions with Update Record actions.
- **Activity 3.2:** Add the Send Email action for the caller.

Milestone 4: Testing and Validation

- **Activity 4.1:** Create a test ticket and verify the Category and Subcategory.
- **Activity 4.2:** Verify the confirmation email in the Emails log.

Milestone 5: Deployment

- **Activity 5.1:** Activate the flow.
- **Activity 5.2:** Export the Update Set as an XML file.

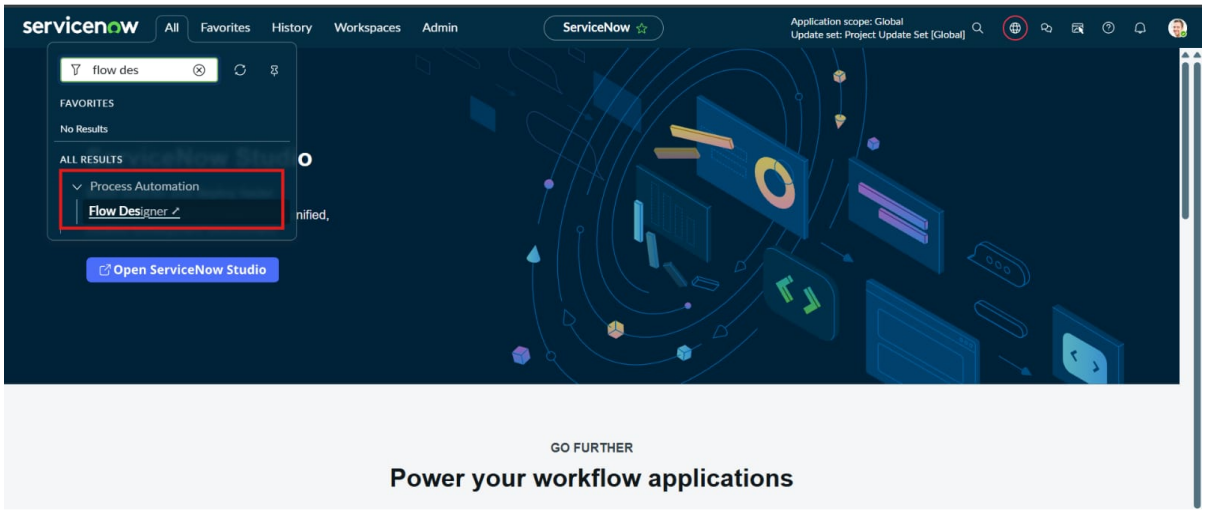
Milestone 6: Conclusion

Milestone 1: Platform Setup and Architecture

In this milestone, we open the ServiceNow environment and decide how tickets will be classified. This involves reaching Flow Designer, identifying the categories and keywords that tickets should be sorted by, and making sure the design matches the objective of classifying tickets and notifying callers automatically.

Activity 1.1: Open Flow Designer

- **Step 1 - Log in:** Open your ServiceNow instance and sign in as an administrator.
- **Step 2 - Search:** In the All menu, type *flow des* in the filter box.
- **Step 3 - Open:** Under *Process Automation*, click **Flow Designer**. It opens in Workflow Studio.



Searching for Flow Designer under Process Automation

Activity 1.2: Define the Classification Rules

Here are the things that need to be decided before configuring the flow:

- **Understand the Requirements:** Review the tickets normally raised in the school or college.
- **Identify the Categories:** Network, Hardware, Access and Performance.
- **Choose the Keywords:** Map each category and subcategory to the words users write in the short description.

Keyword in Short Description	Category	Subcategory
Wi-Fi / Network	Network	Wi-Fi
Projector	Hardware	Projector
Forgot password	Access	Forgot Password
Slow computer	Performance	Slow Computer

Activity 1.3: Define the Architecture of the Application

- **Draft an Architectural Diagram:** Show the Incident WorkFlow form, the Flow Designer flow, the email action and the Update Set.
- **Detail User Interaction:** The caller only enters a short description; no extra input is needed for classification.
- **Outline Automation Responsibilities:** The flow triggers on creation, updates Category and Subcategory and emails the caller.

Activity 1.4: Set Up the Environment

- Confirm the **Incident WorkFlow** table (u_incident_workflow) is available with the fields Caller, Category, Subcategory and Short description.
- Confirm the Category and Subcategory choices (Network / Wi-Fi, Hardware / Projector, Access / Forgot Password, Performance / Slow Computer) exist.
- Make sure the callers have valid email addresses so the notification can be delivered.

Milestone 2: Core Functionalities Development

Activity 2.1: Develop Core Features

Implement the core features as one automated flow that runs whenever a new ticket is created.

- **Ticket Categorizer:** Reads the short description and sets the Category and Subcategory using keyword conditions.
- **Email Notification:** Sends the caller an email with the subject "Your Request for the issue has been submitted."

Activity 2.2: Configure the Data and Email Content

- Use the Trigger data pill *Incident WorkFlow Record* so that every action updates the same ticket that started the flow.
- Prepare the email message: a short greeting from the college saying that the request has been viewed and will be solved in a short time.

Key Points:

- The trigger runs only when the Category is empty, so tickets that are already classified are left unchanged.
- Conditions are checked in order, and the first matching condition decides the Category.

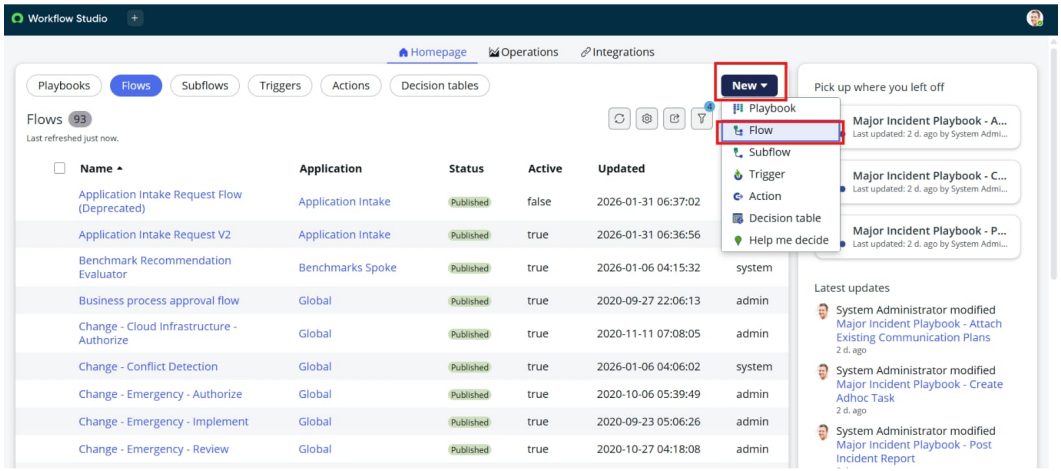
Milestone 3: Flow Designer Development

Milestone 3 focuses on building the main automation in Flow Designer, which is the backbone of the project: the flow, its trigger, the classification conditions and the actions.

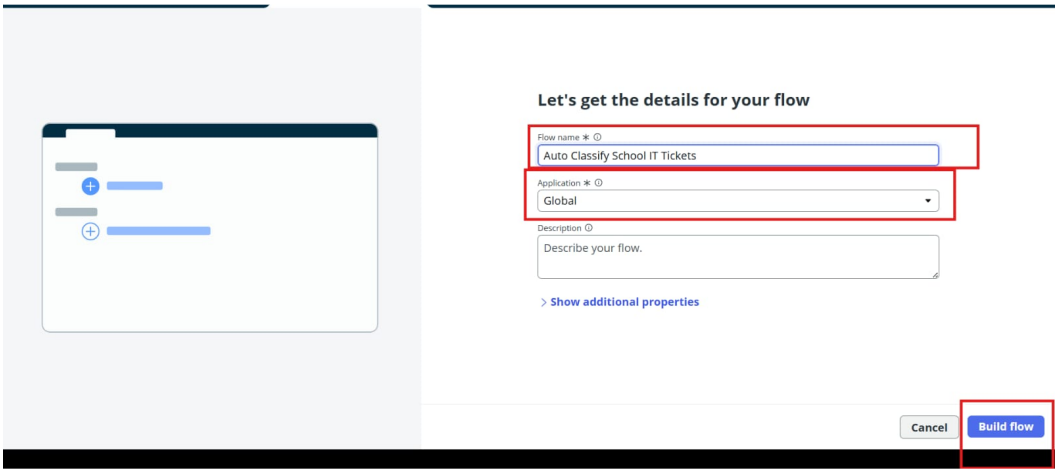
Activity 3.1: Create the Flow and Configure the Trigger

Create the Flow:

- In Workflow Studio, open the **Flows** tab and click **New > Flow**.
- Enter the Flow name **Auto Classify School IT Tickets**, set Application to **Global** and click **Build flow**.



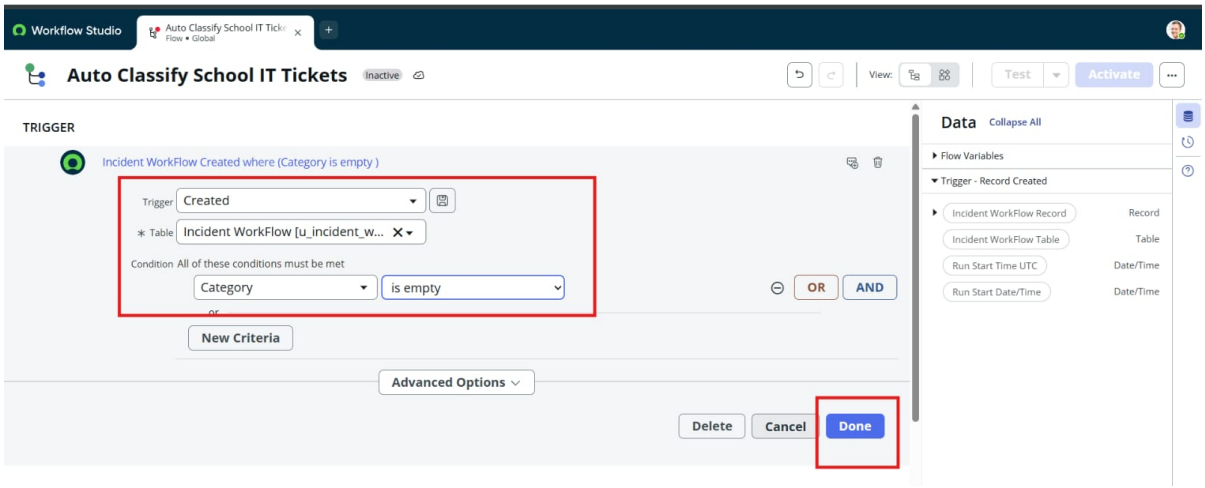
Creating a new flow using New > Flow



Flow name and application details, then Build flow

Configure the Trigger:

- Click **Add a trigger** and choose **Created**.
- Set *Table* to **Incident WorkFlow [u_incident_workflow]** and add the condition **Category is empty**, then click **Done**.



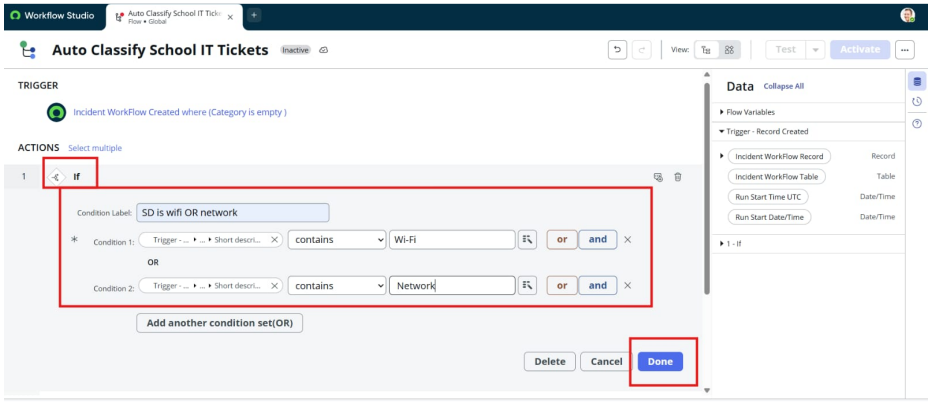
Trigger: record Created on Incident WorkFlow where Category is empty

Add the Classification Conditions and Update Actions:

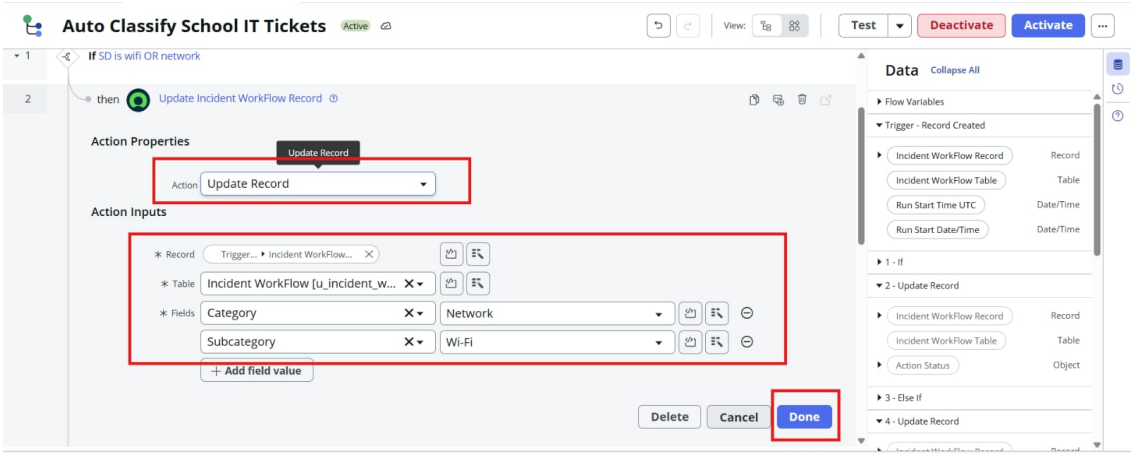
For each keyword, add a Flow Logic block and an Update Record action. The first is an **If** and the others are **Else If** blocks.

1. Network / Wi-Fi

- **If** the Short description *contains* Wi-Fi **OR** Network (condition label "SD is wifi OR network").
- **Then** Update Record: Category = Network, Subcategory = Wi-Fi.



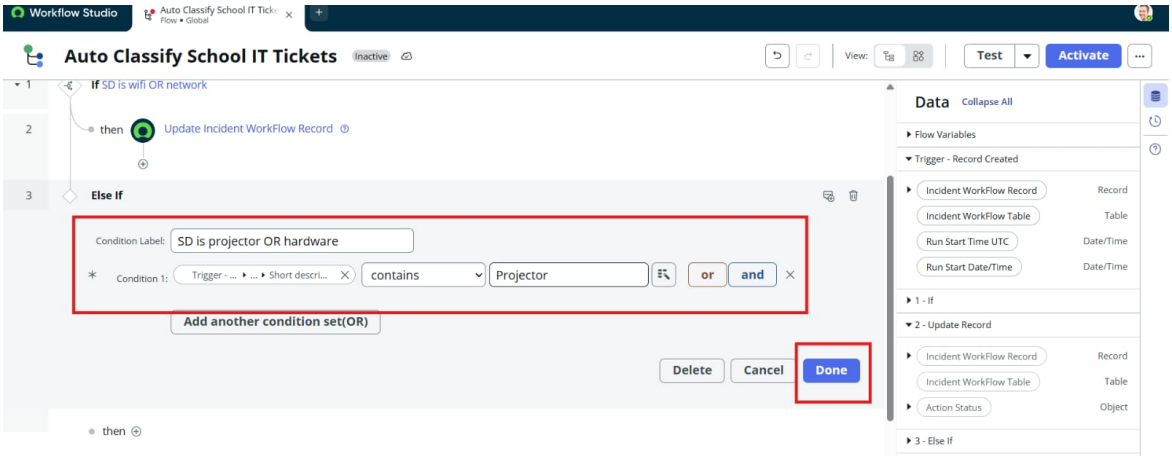
If condition: Short description contains Wi-Fi or Network



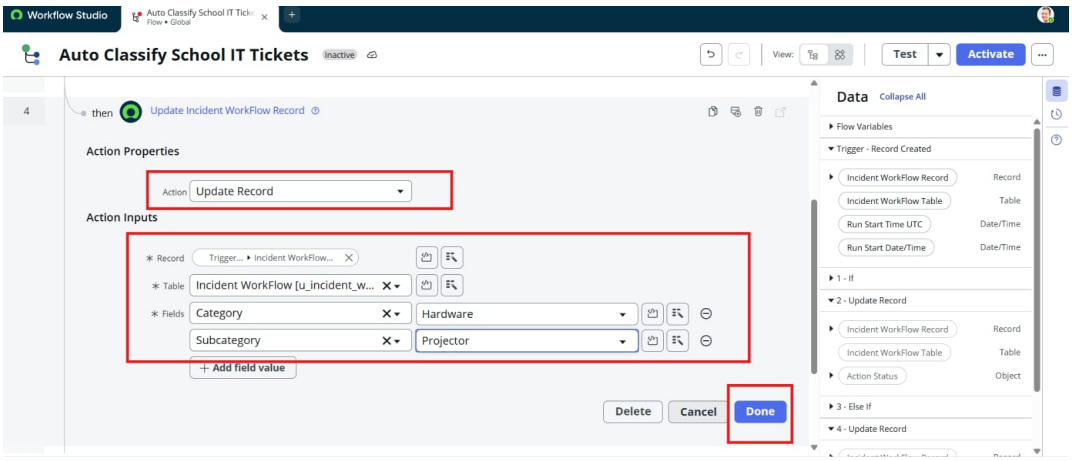
Update Record: Category = Network, Subcategory = Wi-Fi

2. Hardware / Projector

- **Else If** the Short description *contains* Projector (label "SD is projector OR hardware").
- **Then** Update Record: Category = Hardware, Subcategory = Projector.



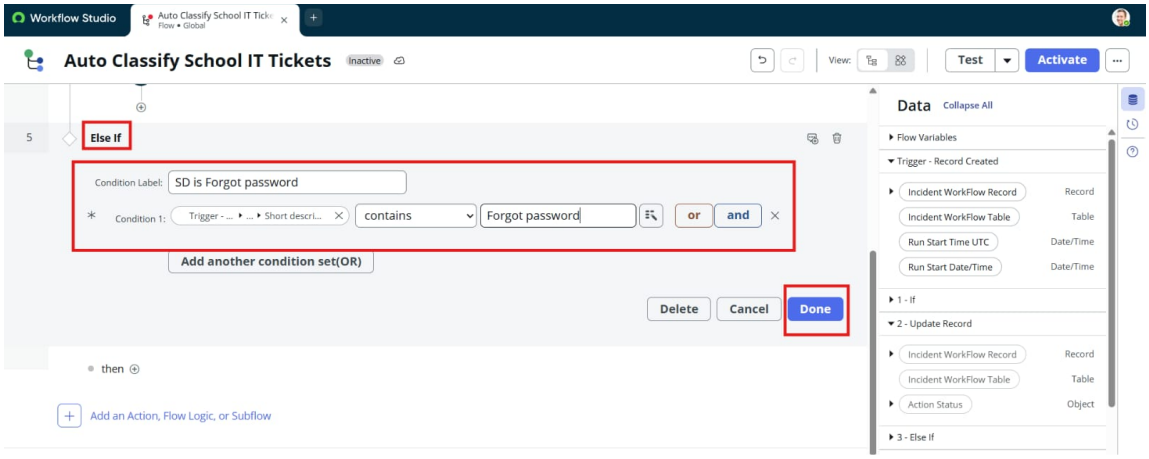
Else If condition: Short description contains Projector



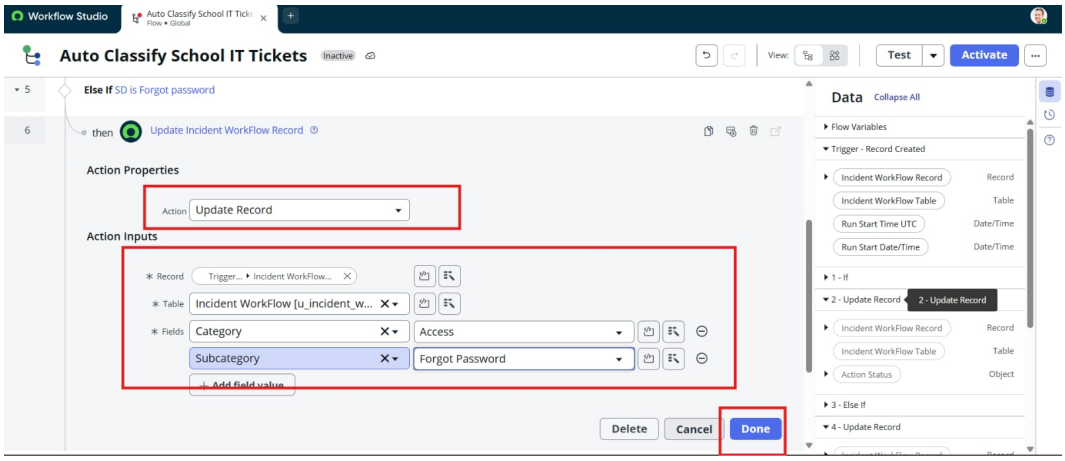
Update Record: Category = Hardware, Subcategory = Projector

3. Access / Forgot Password

- **Else If** the Short description *contains* Forgot password.
- **Then** Update Record: Category = Access, Subcategory = Forgot Password.



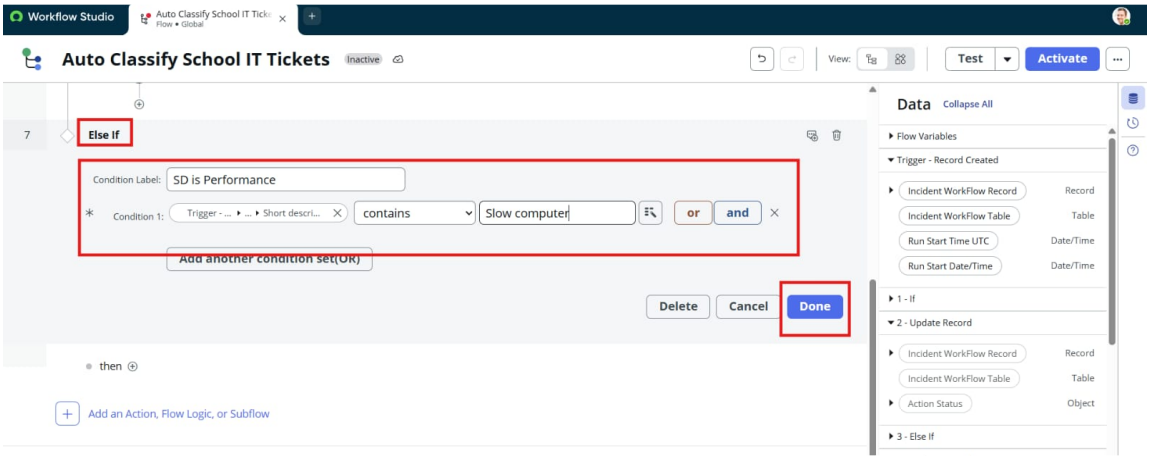
Else If condition: Short description contains Forgot password



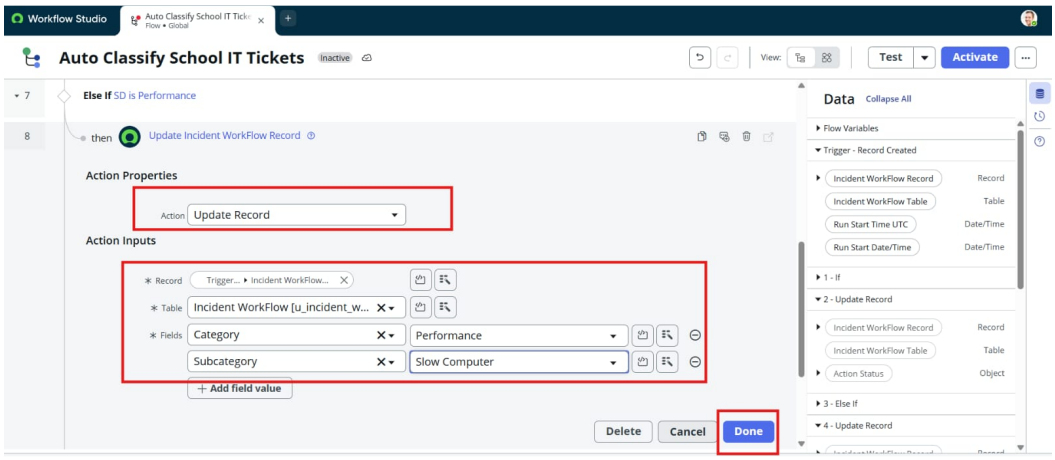
Update Record: Category = Access, Subcategory = Forgot Password

4. Performance / Slow Computer

- **Else If** the Short description *contains* Slow computer (label "SD is Performance").
- **Then** Update Record: Category = Performance, Subcategory = Slow Computer.



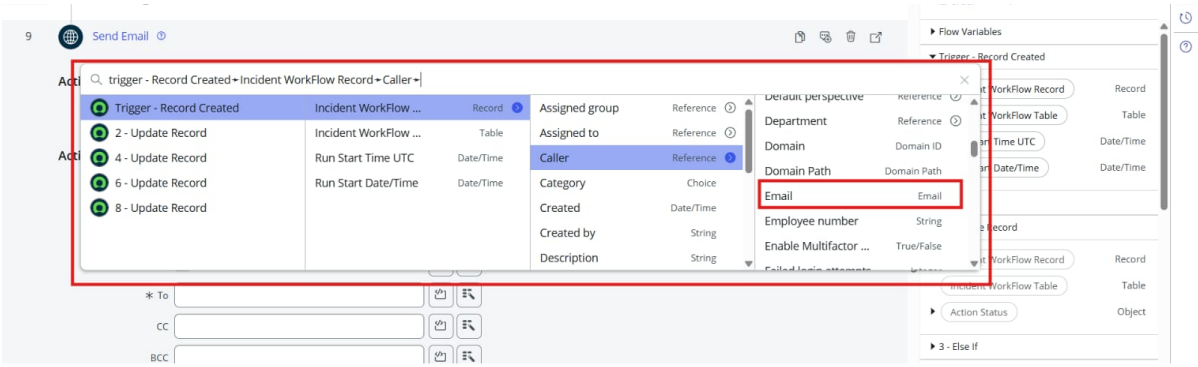
Else If condition: Short description contains Slow computer



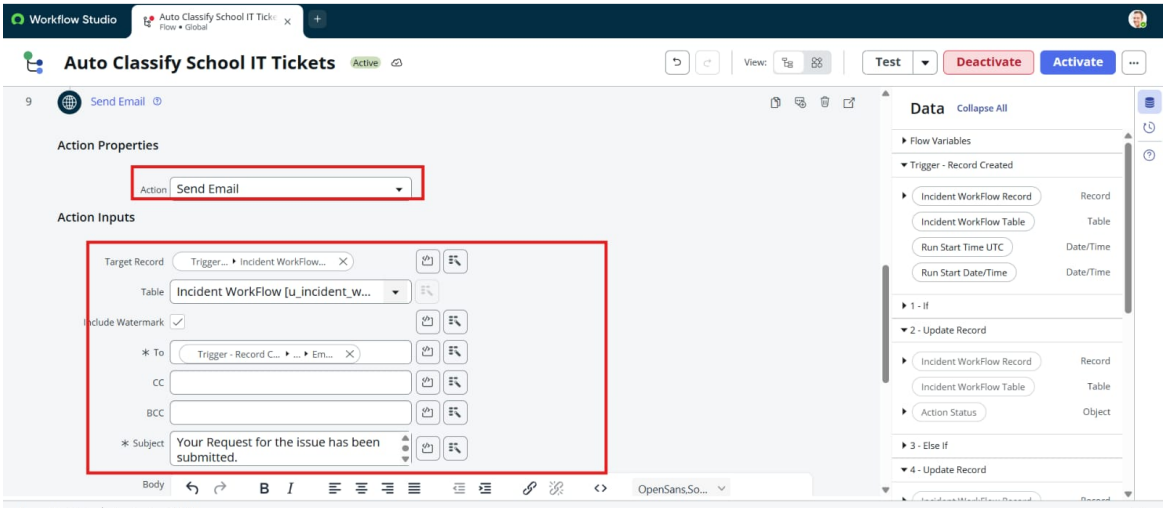
Update Record: Category = Performance, Subcategory = Slow Computer

Activity 3.2: Add the Send Email Action

- Add an action after the conditions and choose **Send Email**.
- Set the **Target Record** to the Trigger Incident WorkFlow record and the **Table** to Incident WorkFlow.
- In the **To** field, use the data pill *Trigger - Record Created > Incident WorkFlow Record > Caller > Email* so the mail goes to the caller.
- Enter the subject **Your Request for the issue has been submitted.** and write the message body.



Selecting Caller > Email as the recipient of the email



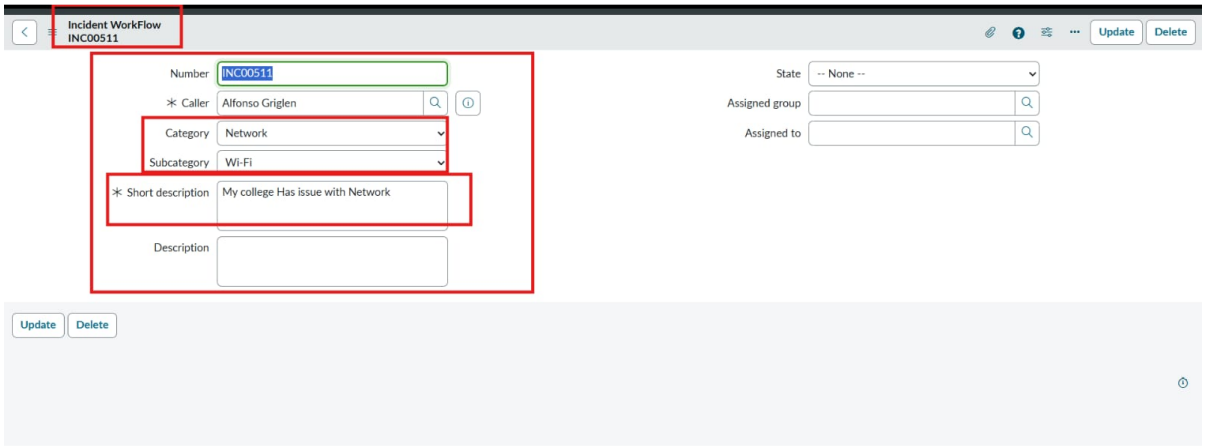
Send Email action with the recipient and subject configured

Milestone 4: Testing and Validation

Milestone 4 focuses on verifying that the flow behaves as expected, by checking both the classified ticket and the confirmation email.

Activity 4.1: Create a Test Ticket

- Open the Incident WorkFlow table and create a new record.
- Select a Caller and enter the Short description *My college Has issue with Network*, leaving Category empty, then click **Submit**.
- Reopen the ticket and check that **Category = Network** and **Subcategory = Wi-Fi** have been filled in automatically.

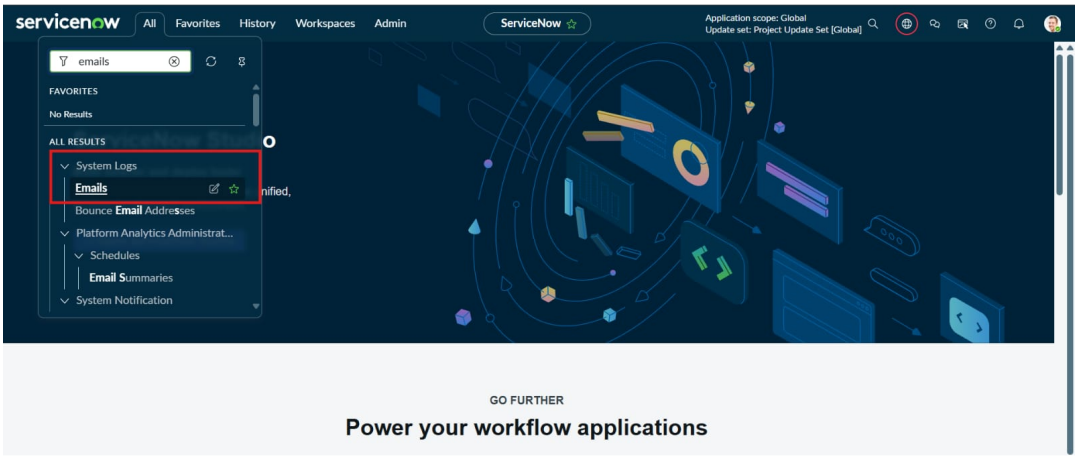


Test ticket INC00511 automatically classified as Network / Wi-Fi

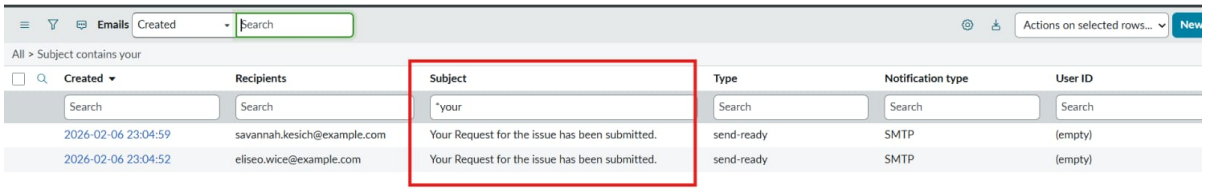
Test Short Description	Expected Result
My college has issue with Network	Network / Wi-Fi
Projector is not working	Hardware / Projector
Forgot password	Access / Forgot Password
Slow computer in lab	Performance / Slow Computer

Activity 4.2: Verify the Confirmation Email

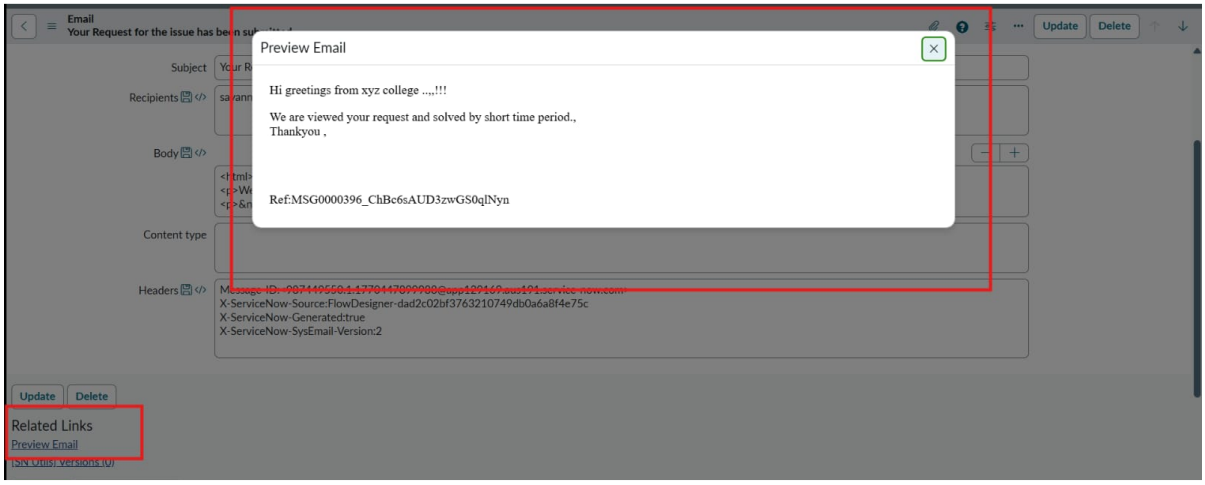
- In the All menu, type *emails* and open **System Logs > Emails**.
- Filter the Subject with *your* to find the message "Your Request for the issue has been submitted."
- Open an email and click **Preview Email** under Related Links to see the message that the caller received.



Opening System Logs > Emails



Emails log showing the confirmation email sent to callers



Preview of the confirmation email

Milestone 5: Deployment

In Milestone 5, the flow is made live on the instance and packaged so it can be shared, imported and verified by others.

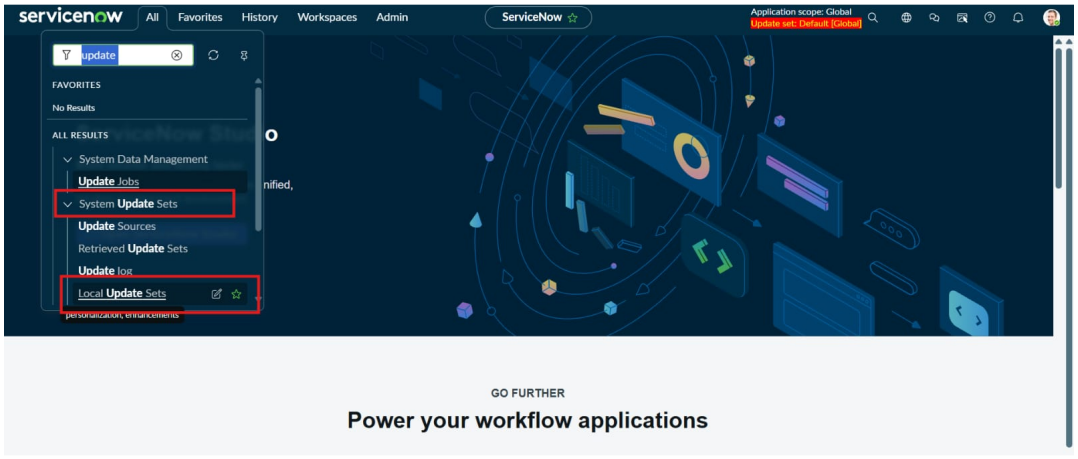
Activity 5.1: Activate the Flow

- Open the flow in Flow Designer and click **Activate**; the status changes from Inactive to **Active**.
- Create a fresh ticket and confirm it is classified automatically.

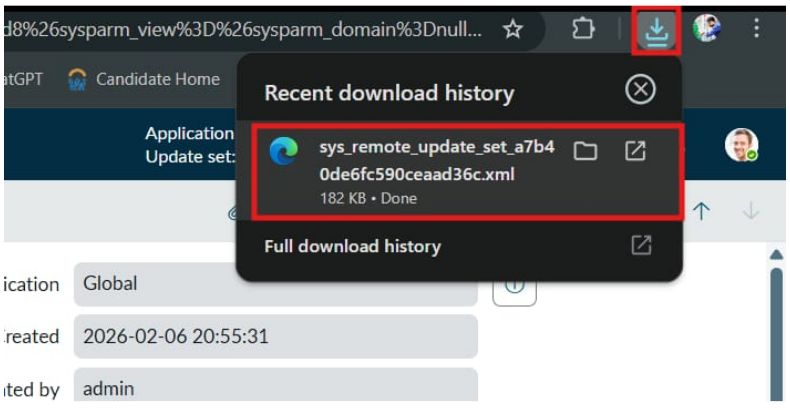
Activity 5.2: Export the Update Set

An Update Set captures all the changes made during development so the solution can be moved to another instance.

- In the All menu, type *update* and open **System Update Sets > Local Update Sets**.
- Open the update set used for this project and export it using **Export to XML**.
- The browser downloads an XML file such as *sys_remote_update_set_a7b40de6fc590ceaad36c.xml*.



Opening System Update Sets > Local Update Sets



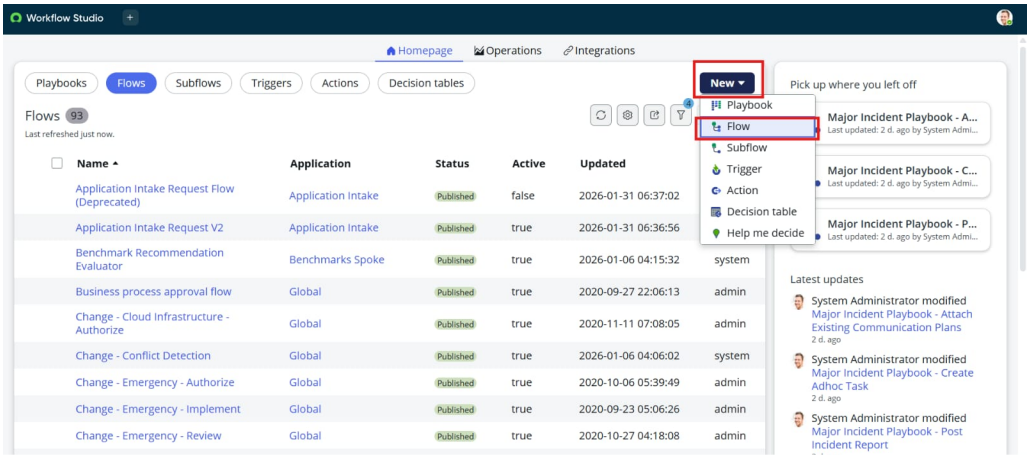
Update Set XML file downloaded (182 KB)

Important Notes:

- A Personal Developer Instance may hibernate when idle; wake it before testing or demonstrating.
- Only newly created tickets with an empty Category are classified by this flow.

Exploring the Application Pages:

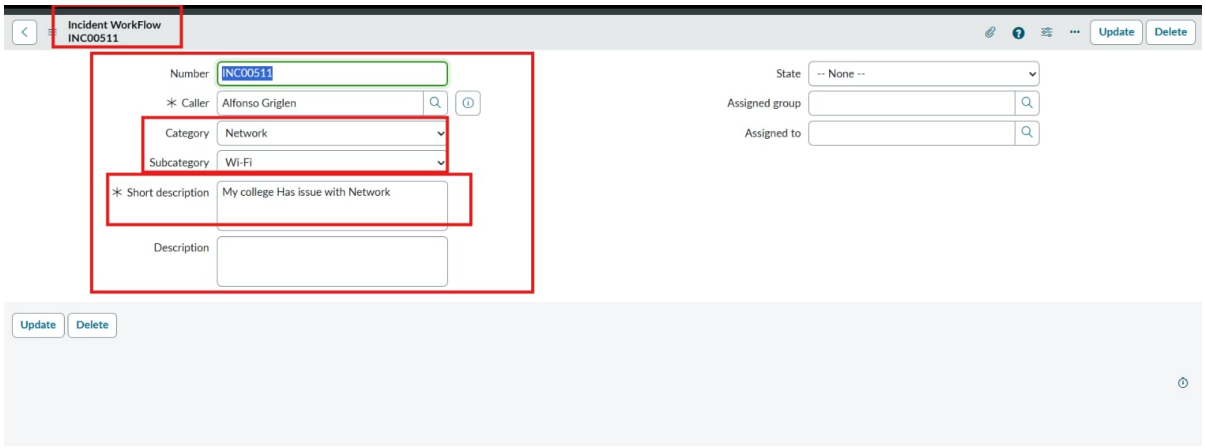
Workflow Studio Homepage:



Workflow Studio - Flows list with the New menu

Description: The Workflow Studio homepage lists all flows on the instance and is the starting point for creating the Auto Classify School IT Tickets flow.

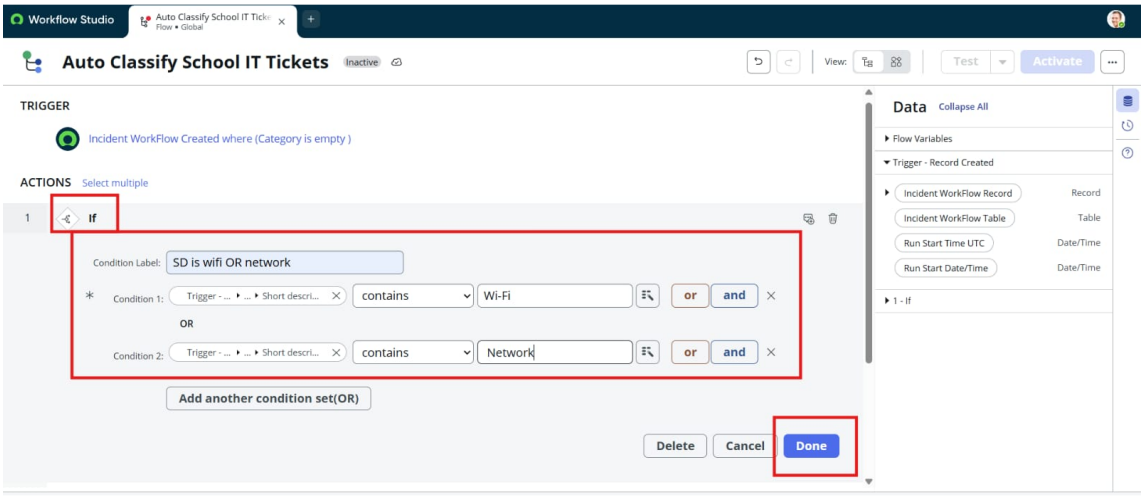
Incident WorkFlow Form:



Incident WorkFlow record after automatic classification

Description: The ticket form shows the Caller, Category, Subcategory and Short description. After submission, the Category and Subcategory are filled in by the flow.

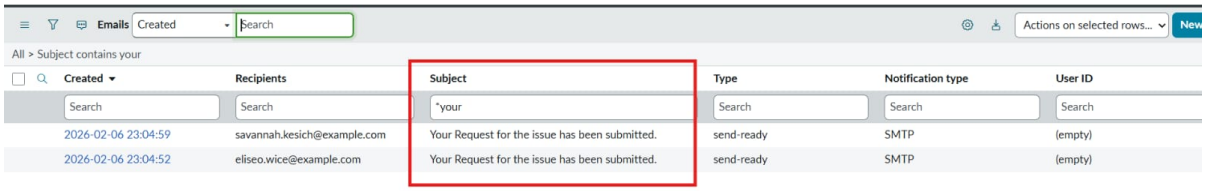
Flow Designer Canvas:



Flow with the trigger and the first If condition

Description: Shows the Created trigger on the Incident WorkFlow table and the If / Else If conditions that check the short description before updating the ticket.

Emails Log:



System Logs > Emails

Description: Lists the confirmation emails sent by the flow, which helps in verifying and troubleshooting notifications.

Conclusion:

Auto Ticket Classification using Flow Designer is a low-code automation solution built on the ServiceNow Now Platform that removes manual sorting of IT support tickets in a school or college. By reading the short description of each new ticket, it fills in the Category and Subcategory and sends the caller a confirmation email, so tickets are organised faster and callers know that their request has been received. The project, which covered platform setup, flow development, testing and deployment through an Update Set, shows how a structured automation can improve support work with no custom code. Looking ahead, the solution can be extended with more keyword rules, automatic assignment groups and priority settings, Predictive Intelligence for smarter classification, issue-specific email messages and multi-language support. By continuously refining the rules, the solution can grow from a simple classification tool into a complete ticket management assistant.